In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

iris = load_iris()

data = pd.DataFrame(
    iris.data,
    columns=[
        'SepalLengthCm',
        'SepalWidthCm',
        'PetalLengthCm',
        'PetalWidthCm'
    ]
)

data['variety'] = pd.Categorical.from_codes(
    iris.target,
    iris.target_names
).map({
    'setosa': 'Iris-setosa',
    'versicolor': 'Iris-versicolor',
    'virginica': 'Iris-virginica'
})

data.insert(0, 'Id', range(1, len(data) + 1))

data.head()
```

Out[1]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [2]:
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype   
---  ------         --------------  -----   
 0   Id             150 non-null    int64   
 1   SepalLengthCm  150 non-null    float64 
 2   SepalWidthCm   150 non-null    float64 
 3   PetalLengthCm  150 non-null    float64 
 4   PetalWidthCm   150 non-null    float64 
 5   variety        150 non-null    category
dtypes: category(1), float64(4), int64(1)
memory usage: 6.2 KB
```

In [3]:
```
data.describe()
```

Out[3]:

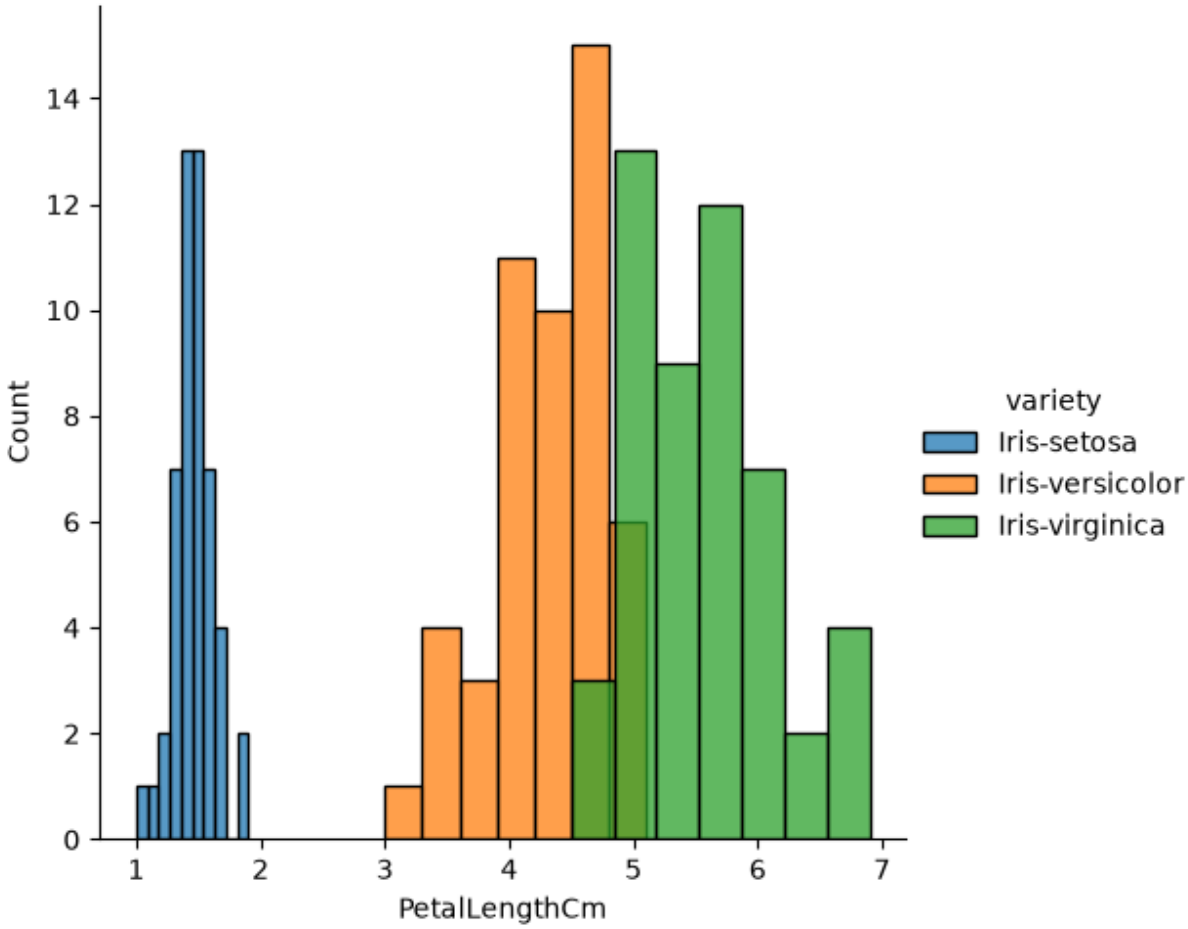
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.057333	3.758000	1.199333
std	43.445368	0.828066	0.435866	1.765298	0.762238
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

In [4]:
```
data.value_counts('variety')
```

Out[4]:
```
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

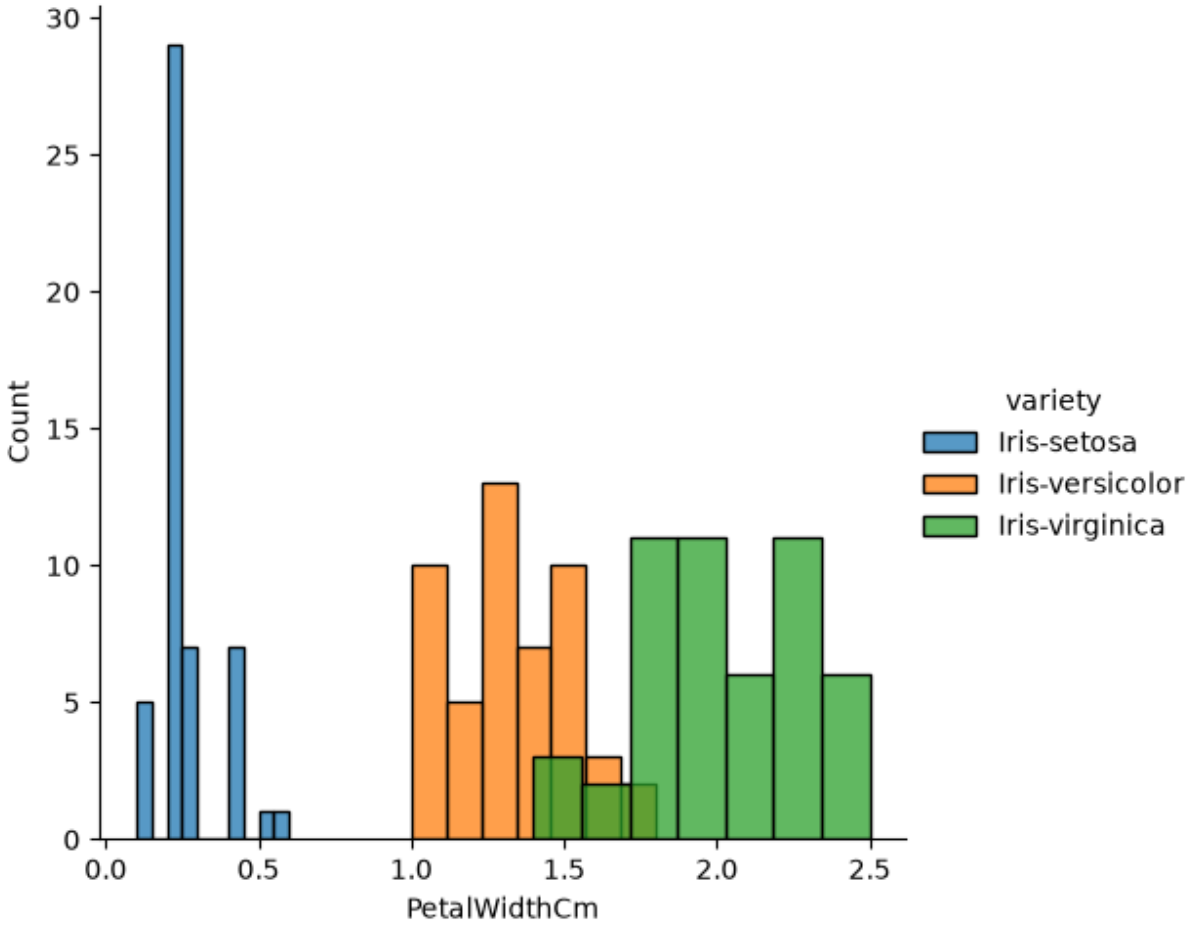
In [5]:
```
sns.FacetGrid(data, hue='variety', height=5).map(
    sns.histplot, 'PetalLengthCm'
).add_legend()

plt.show()
```



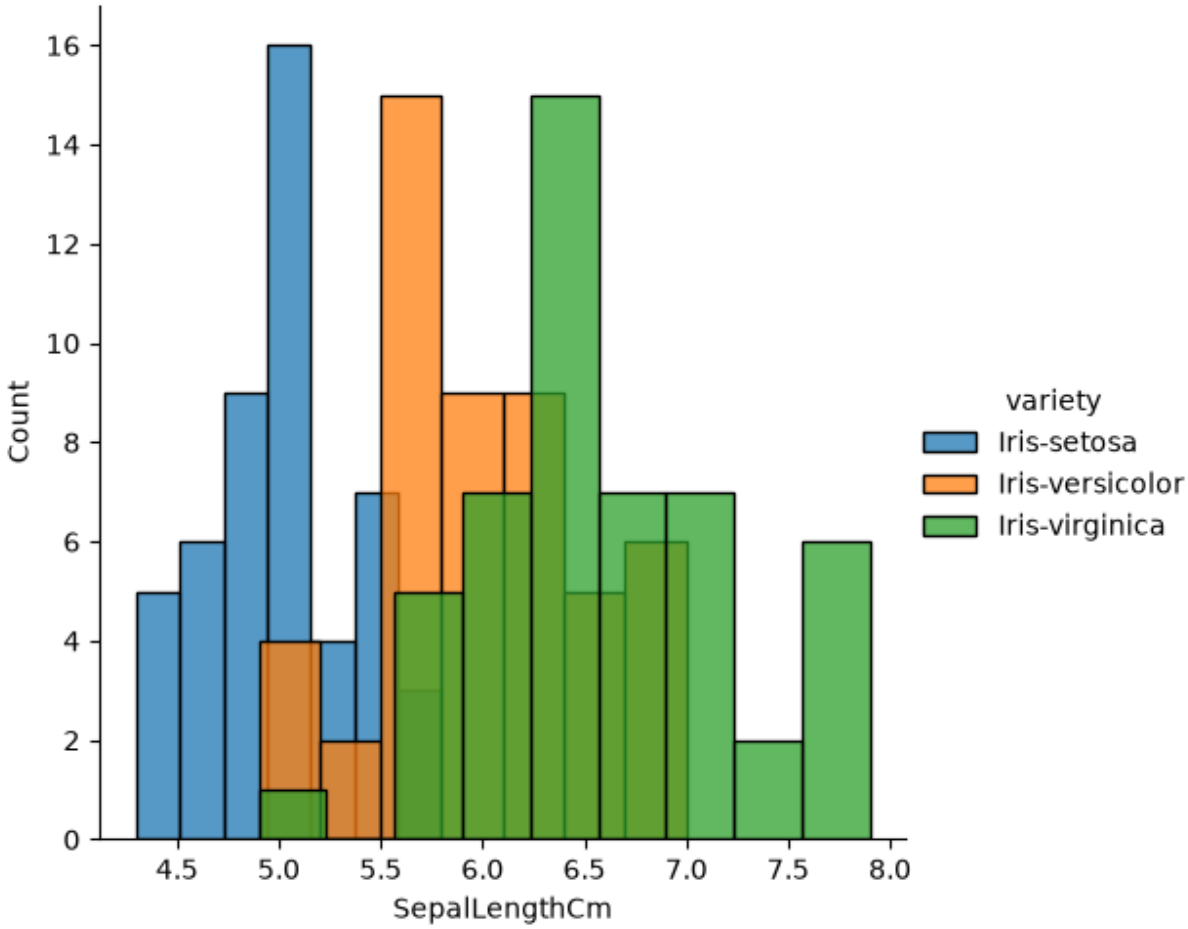
In [6]:
```
sns.FacetGrid(data, hue='variety', height=5).map(
    sns.histplot, 'PetalWidthCm'
).add_legend()

plt.show()
```



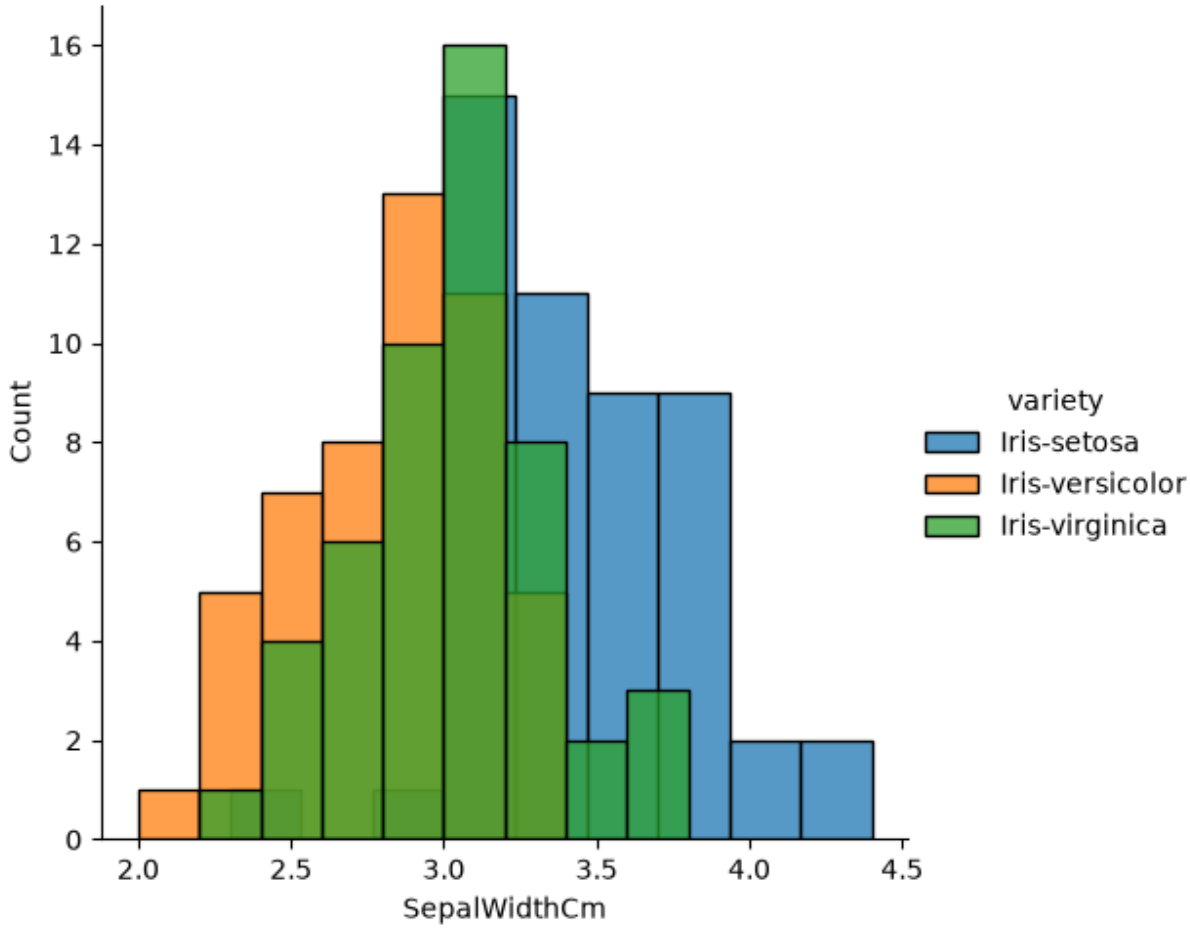
In [7]:
```
sns.FacetGrid(data, hue='variety', height=5).map(
    sns.histplot, 'SepalLengthCm'
).add_legend()

plt.show()
```



In [8]:
```
sns.FacetGrid(data, hue='variety', height=5).map(
    sns.histplot, 'SepalWidthCm'
).add_legend()

plt.show()
```

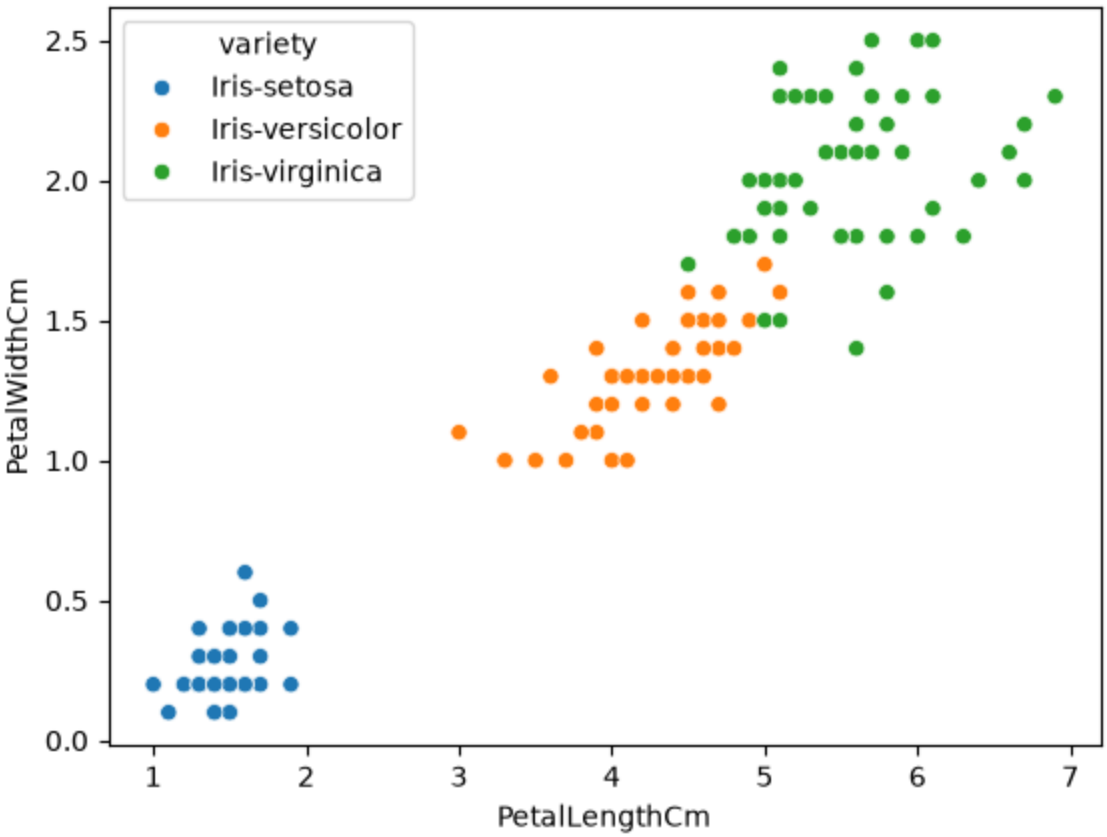


In [9]:
```
%pip install scikit-learn seaborn
```

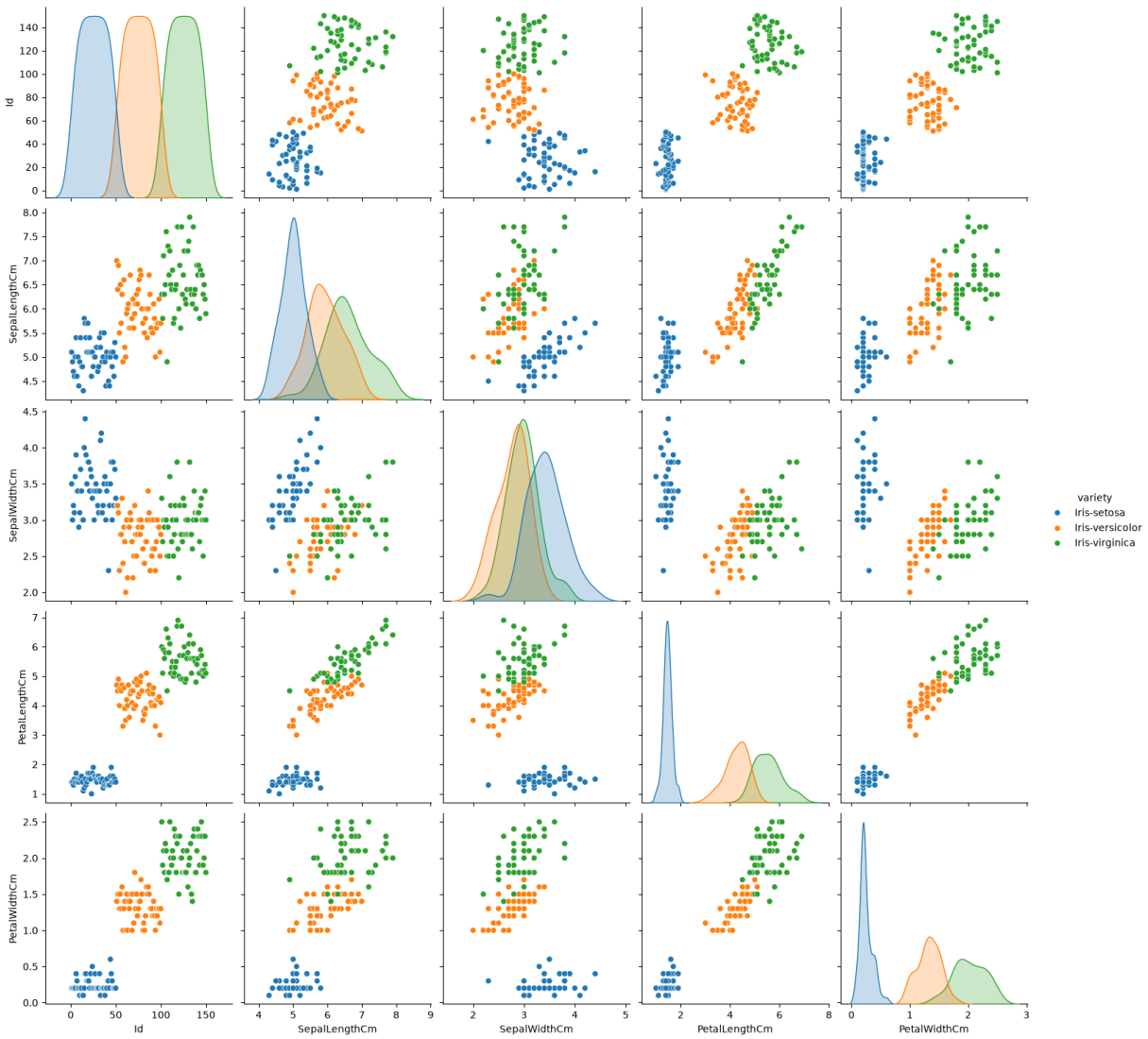
```
Requirement already satisfied: scikit-learn in .\AppData\Local\Python\pythoncore-3.1
4-64\Lib\site-packages (1.9.1)
Requirement already satisfied: seaborn in .\AppData\Local\Python\pythoncore-3.14-64
\Lib\site-packages (0.13.2)
Requirement already satisfied: numpy>=1.24.1 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from scikit-learn) (2.5.3)
Requirement already satisfied: scipy>=1.10.0 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from scikit-learn) (1.18.1)
Requirement already satisfied: joblib>=1.4.0 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from scikit-learn) (2.26.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in .\AppData\Local\Python\python
core-3.14-64\Lib\site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: pandas>=1.2 in .\AppData\Local\Python\pythoncore-3.1
4-64\Lib\site-packages (from seaborn) (3.0.6)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in .\AppData\Local\Python\pyt
honcore-3.14-64\Lib\site-packages (from seaborn) (3.11.2)
Requirement already satisfied: cloudpickle>=3.0 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from joblib>=1.4.0->scikit-learn) (3.1.2)
Requirement already satisfied: contourpy>=1.0.1 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.0)
Requirement already satisfied: cycler>=0.10 in .\AppData\Local\Python\pythoncore-3.1
4-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.66.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.5.1)
Requirement already satisfied: packaging>=20.0 in .\AppData\Local\Python\pythoncor
e-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.3)
Requirement already satisfied: pillow>=9 in .\AppData\Local\Python\pythoncore-3.14-6
4\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.3.0)
Requirement already satisfied: pyparsing>=3 in .\AppData\Local\Python\pythoncore-3.1
4-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in .\AppData\Local\Python\python
core-3.14-64\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: tzdata in .\AppData\Local\Python\pythoncore-3.14-64\L
ib\site-packages (from pandas>=1.2->seaborn) (2026.4)
Requirement already satisfied: six>=1.5 in .\AppData\Local\Python\pythoncore-3.14-64
\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn)
(1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [10]:

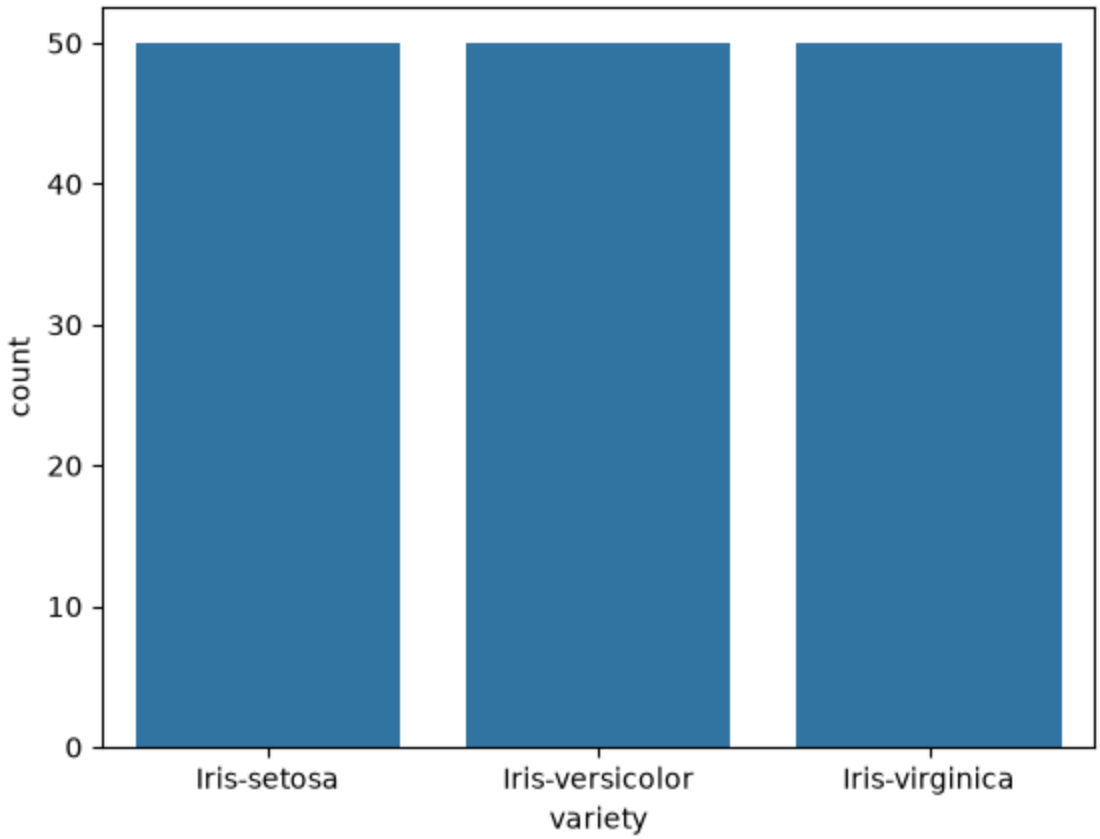
```
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)
plt.show()
```



In [11]:
```
sns.pairplot(data, hue='variety', height=3)
plt.show()
```



In [12]:
```
sns.countplot(x='variety', data=data)
plt.show()
```



In [13]:
```
dummies = pd.get_dummies(data['variety'])
dummies
```

Out[13]:

	Iris-setosa	Iris-versicolor	Iris-virginica
0	True	False	False
1	True	False	False
2	True	False	False
3	True	False	False
4	True	False	False
...	...	...	...
145	False	False	True
146	False	False	True
147	False	False	True
148	False	False	True
149	False	False	True

150 rows × 3 columns

In [14]:
```
FinalDataset = pd.concat(
```

```
    [dummies, data.iloc[:, [0, 1, 2, 3]]],
    axis=1
)

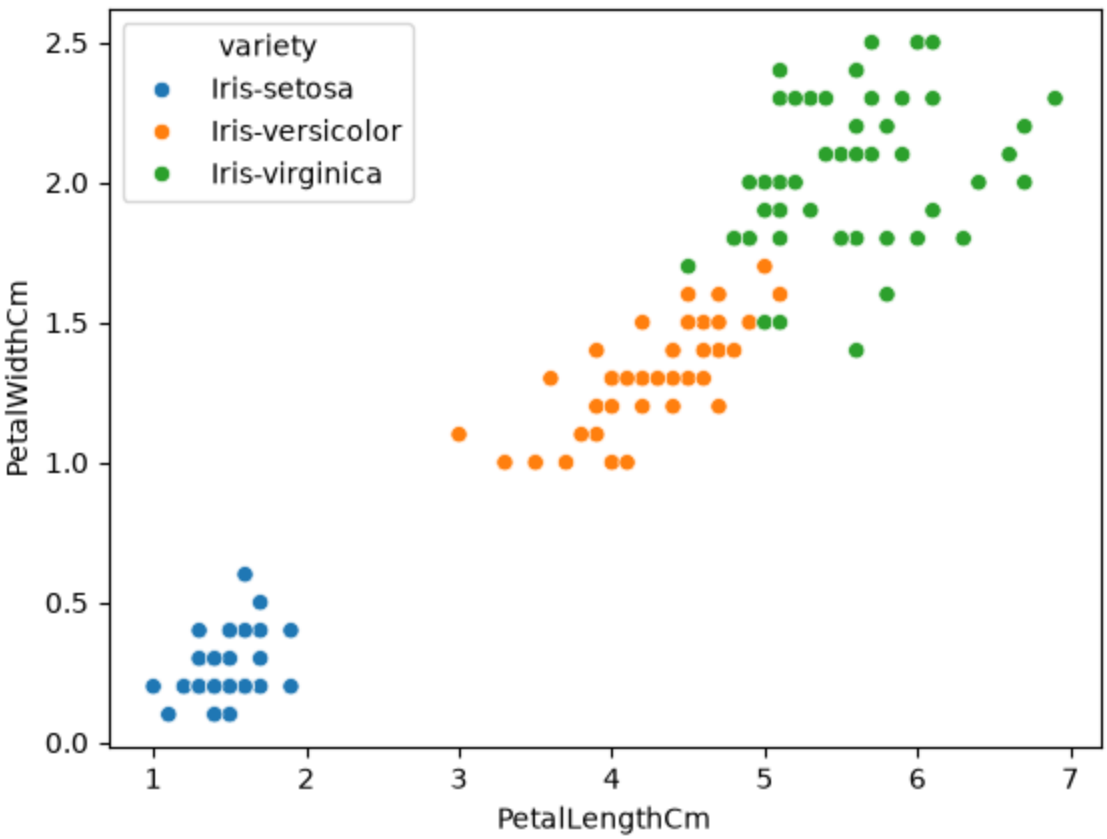
FinalDataset.head()
```

Out[14]:

	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

In [16]:

```
sns.scatterplot(
    x='PetalLengthCm',
    y='PetalWidthCm',
    hue='variety',
    data=data
)
plt.show()
```



In [ ]: